

Doubly Fed Induction Generator Subject to Symmetrical Voltage Sags

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Abstract—The aim of this paper is to analyze the dynamic behavior of the doubly fed induction generator (DFIG) subject to symmetrical voltage sags caused by three-phase faults. A simple control algorithm is considered and assumed ideal: the rotor current i_{rf} in the synchronous reference frame is kept constant. This hypothesis allows the electrical transient to be solved analytically, providing a comprehensive description of DFIG behavior under symmetrical sags. The fault-clearing physics of symmetrical sags is also analyzed. That is, the fault is cleared in the successive natural fault-current zeros, leading to a voltage recovery in one, two, or three steps. This clearing process, called *discrete* fault clearing in this paper, results in a more accurate sag modeling than the *abrupt* or instantaneous fault clearing (the usual modeling in the literature). The fault-clearing process has a strong influence on the rotor voltage required to control the rotor current after fault clearing. To compare the effects of both *abrupt* and *discrete* sags, different wind turbine (WT) operating points, which determine different generated powers, are considered. This study helps in the understanding of WT fault ride-through capability.

Index Terms—Doubly fed induction generator (DFIG), fault clearing, fault ride-through capability, symmetrical voltage sag.

I. INTRODUCTION

THE most commonly used concept for variable-speed wind turbines (WTs) is based on the doubly fed induction generator (DFIG) [1]. Fig. 1 shows the basic scheme of this configuration, whose main advantage is that the converter handles a fraction (about 30%) of the total generated power.

In order to meet current grid codes, variable-speed WTs must achieve fault ride-through capability, i.e., they must remain connected to the electrical grid when faults occur. Consequently, knowledge of the effects of voltage sags on DFIG dynamic behavior is the first step to ensure electricity supply continuity.

Although DFIG behavior under voltage sags has been widely studied in the literature, only few studies focus on sizing the rotor-side converter to overcome grid faults and enhancing fault ride-through capability. For example, Chondrogiannis and Barnes [2] present an analytical study to size the rotor-side

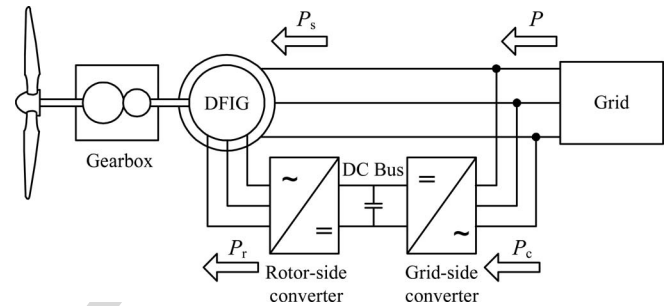


Fig. 1. Scheme of a DFIG-based WT. Motor sign convention.

converter to improve symmetrical sag ride-through capability assuming that the active and reactive powers are controlled during the fault, and Hu and He [3] study the unsymmetrical sag effects considering the rotor-side converter limits.

The analytical approach in this study is based on the assumption that the rotor current i_{rf} in the synchronous reference frame can be kept constant during and after the fault. It allows the electrical transient to be solved analytically, providing a comprehensive description of DFIG behavior. The voltage limit of the rotor-side converter is considered to discuss if the rotor current can be effectively controlled. The analytical expressions were validated by the numerical integration of the system equations with a MATLAB ode solver and the PSpice circuit simulator.

Furthermore, two approaches for the modeling of sags caused by faults are compared in this paper. The first one, known as *abrupt* recovery sag, is the widely accepted model in the technical literature, where it is assumed that the voltage is abruptly recovered in all faulted phases. The second one, i.e., *discrete* recovery sag, considers the fault-clearing physics in order to recover the voltage in different steps depending on the number of faulted phases [4]. The first studies in the literature about the influence of fault clearing were carried out in [5]–[7] for three-phase transformers and induction motors, showing that the fault-clearing process has a strong influence on the behavior of such devices, smoothing the sag effects. It is also true in DFIG behavior under symmetrical sags, where the *abrupt* recovery sags overestimate the severity of sag events, as shown in this study.

Finally, the influence of the generated power, i.e., the WT operating point influence, is also taken into account to compare the effects caused by symmetrical sags.

II. VOLTAGE SAG CHARACTERIZATION

A voltage sag is defined as a decrease in the rms voltage between 0.1 and 0.9 pu for durations from 0.5 cycles to 1 min [8].

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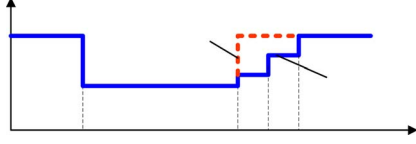


Fig. 2. Evolution of the rms voltage during a symmetrical sag caused by a three-phase-to-ground fault. Dotted line: sag modeled with *abrupt* voltage recovery, or type A sag. Solid line: sag modeled with three-step voltage recovery (*discrete*), or type A₃ sag.

This paper deals with voltage sags caused by three-phase faults (symmetrical voltage sags).

Voltage sags are characterized by four parameters [9], namely, duration Δt , depth h , fault current angle ψ , and typology. Moreover, no phase-angle jump is considered.

In this paper, two approaches for the modeling of sags caused by faults are compared. The first one, the most common in the literature, assumes that faults are cleared instantaneously in all affected phases, i.e., voltage recovery is modeled *abrupt*. This approach is used in Sections VI and VII. However, voltage recovery is influenced by the fault-clearing physics because the voltage recovers in the natural fault-current zeros. This leads to a voltage recovery in one, two, or three steps, that is, to a *discrete* recovery sag [4]. This approach is used in Section VIII. Fig. 2 shows the rms voltage waveform pattern of a sag caused by a three-phase-to-ground fault modeled with *abrupt* and *discrete* voltage recovery. t_{f1} is the instant when the first phase of the circuit breaker clears the fault, and t_{f2} and t_{f3} are the instants when the second and third phases of the circuit breaker clear the fault. Finally, both models (*abrupt* and *discrete*) are compared in Section IX.

III. DFIG DYNAMIC MODEL

In the literature, DFIG dynamic equations are usually written in *Park* variables (dq components) [10]. In this paper, the *Ku* transformation [11], [12] is preferred because the equations are written in a more compact form as the *forward* component (f component) is a complex notation of the dq components [12]. The relation between the *Park* and *Ku* components is given in Appendix I.

When using the *Ku* transformation in the synchronous reference frame and considering the motor sign convention, the DFIG dynamic equations are

$$\begin{aligned} v_{sf} &= [R_s + L_s(p + j\omega_s)] i_{sf} + M(p + j\omega_s) i_{rf} \\ v_{rf} &= [R_r + L_r(p + j\omega_s)] i_{rf} + M(p + j\omega_s) i_{sf} \\ \Gamma_m &= 2\varphi M \text{Im}(i_{sf} i_{rf}^*) \quad s = \frac{(\omega_s - \varphi\omega_m)}{\omega_s} \end{aligned} \quad (1)$$

where subscripts s and r stand for the stator and the rotor, respectively, subscript f stands for the *forward* component of the transformed variable, φ is the number of pole pairs, p is the time differential operator d/dt , Γ_m is the electromagnetic torque, s is the mechanical slip, ω_m is the generator speed, and $\omega_s = 2\pi f_s$ is the pulsation of the stator voltages (f_s is the frequency of the stator voltages, and $T = 1/f_s$ is the period).

TABLE I
2 MW DFIG-BASED WT CHARACTERISTICS (ADAPTED FROM [13])

GENERATOR NOMINAL VALUES				GENERATOR SPEED		POLE PAIRS	
U_n (line voltage)		f_n		ω_m		p	
690 V		50 Hz		900-1900 rpm		2	
GENERATOR PARAMETERS IN PU ($S_b = 2$ MW, $U_b = 690$ V, $f_b = 50$ Hz)							
R_s	R_r	X_{sl}	X_{rl}	X_m	Inertia constant		
0.01	0.01	0.1	0.08	3.0	0.5 s		
WIND TURBINE PARAMETERS							
Nominal power	Blade radius	Min. speed	Max. speed	Nominal speed	Nominal wind speed	Gear box ratio	Inertia constant
2 MW	37.5 m	9 rpm	19 rpm	18 rpm	12 m/s	1:100	2.5 s

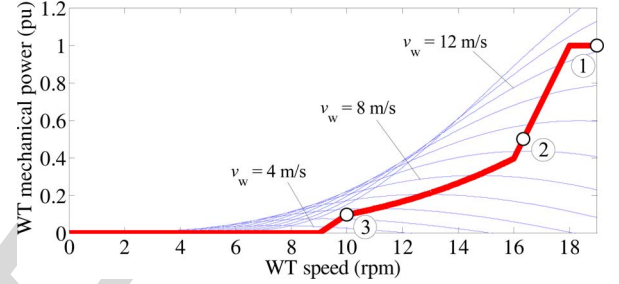


Fig. 3. Steady-state WT mechanical power (thick line) for different values of the wind speed (thin lines). 1, 2, and 3 are the operating points studied in the paper.

The total instantaneous power $p(t)$ exchanged between the WT and the grid is obtained as

$$\begin{aligned} p(t) &= p_s(t) + p_c(t) = p_s(t) + p_r(t) \\ &= 2 \text{Re}(v_{sf} i_{sf}^*) + 2 \text{Re}(v_{rf} i_{rf}^*) \end{aligned} \quad (2)$$

where the converter losses are neglected ($p_c = p_r$, Fig. 1).

IV. DFIG-BASED WT CHARACTERISTICS

The 2 MW DFIG-based WT characteristics are shown in Table I. This educational data, as well as the coefficients c_p used to obtain the power coefficient of the WT, were adapted from [13]. The resulting steady-state WT mechanical power versus the WT speed for different wind speeds is represented in Fig. 3. The influence of the total generated power P (which depends on the wind speed, v_w) on DFIG behavior is discussed in Section IX, where the three operating points in Fig. 3 are studied:

- 1) Point 1: rated power ($v_w = 12$ m/s, $s = -0.27$).
- 2) Point 2: 0.5 times the rated power ($v_w = 9.3$ m/s, $s = -0.09$).
- 3) Point 3: 0.1 times the rated power ($v_w = 5.45$ m/s, $s = 0.33$).

V. DFIG VARIABLES OF INTEREST

The variables of interest are the abc phase components and the *forward* components of the voltages and currents, as well as the total instantaneous power. The pu values of these variables are obtained by referring them to the base values ($S_b = 2$ MW,

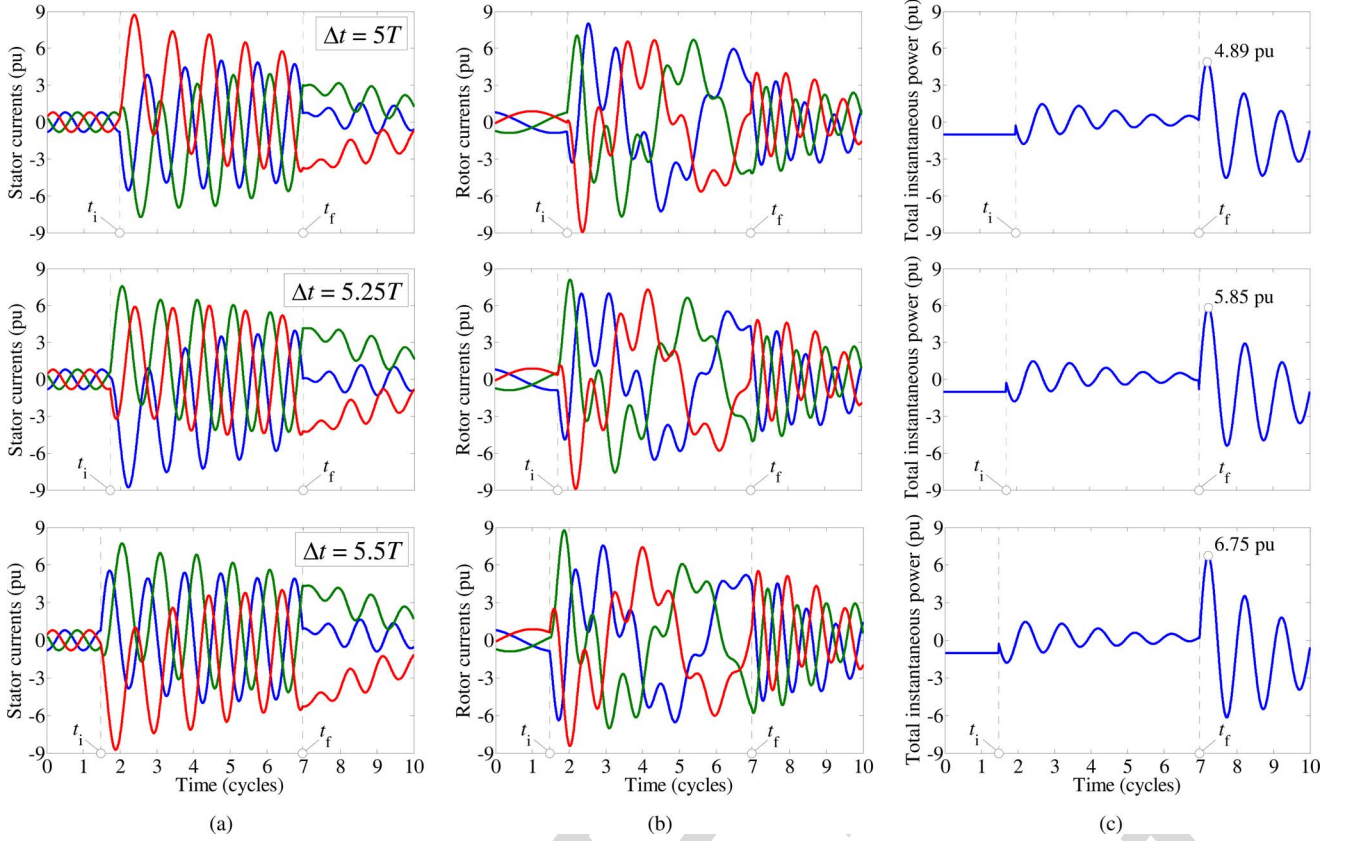


Fig. 4. Influence of sag duration ($\Delta t = 5T$, $\Delta t = 5.25T$, and $\Delta t = 5.5T$) on the dynamic behavior of the DFIG with no control on the rotor current. *Abrupt* recovery sag characteristics: $h = 0.1$ and $\psi = 80^\circ$. WT operating point 1: P_n and $s = -0.27$. (a) Stator currents, (b) rotor currents, and (c) total instantaneous power.

137 $U_b = 690 \text{ V}$, $I_b = S_b / (\sqrt{3}U_b) = 1673.5 \text{ A}$:

$$\begin{aligned}
 v_{abc \text{ pu}}(t) &= \frac{v_{abc}(t)}{\sqrt{2}V_b} & v_{f \text{ pu}}(t) &= \frac{v_f(t)}{\sqrt{3/2}V_b} \\
 i_{abc \text{ pu}}(t) &= \frac{i_{abc}(t)}{\sqrt{2}I_b} & i_{f \text{ pu}}(t) &= \frac{i_f(t)}{\sqrt{3/2}I_b} & p_{\text{pu}}(t) &= \frac{p(t)}{S_b}
 \end{aligned}
 \quad (3)$$

138 where $V_b = U_b / \sqrt{3} = 398.4 \text{ V}$ is the per-phase base voltage
 139 and $\sqrt{3/2}$ is the relation between the steady-state *forward* com-
 140 ponents in the synchronous reference frame and the steady-state
 141 phasors (see Appendix I for more details). In the remainder of
 142 this paper, subscript pu is omitted for brevity.

143 VI. DFIG WITH NO CONTROL ON ROTOR CURRENT UNDER 144 SYMMETRICAL SAGS WITH *Abrupt* RECOVERY

145 Fig. 4 illustrates the dynamic behavior of the DFIG stator
 146 currents [see Fig. 4(a)], rotor currents [see Fig. 4(b)], and total
 147 instantaneous power [see Fig. 4(c)] when subject to a symmet-
 148 rical sag with *abrupt* recovery and three different durations:
 149 $\Delta t = 5T$, $\Delta t = 5.25T$, and $\Delta t = 5.5T$. There is no control on
 150 the rotor current (i.e., the transformed rotor voltage v_{rf} in the
 151 synchronous reference frame is constant and equals the prefault

steady-state value) and the WT operates at point 1 (P_n and $s =$ 152
 -0.27). Two observations can be made from the figure results. 153

- 1) The rotor current peaks that appear in a DFIG subject to 154
 symmetrical sags cannot be tolerated by the converter 155
 (the peaks are 9 pu in the examples). Therefore, a control 156
 strategy on the rotor current is needed to avoid or mitigate 157
 these peaks. 158
- 2) The most severe peaks in the total instantaneous power 159
 occur when sag duration is $\Delta t = 5.5T$, while the least 160
 severe peaks appear for $\Delta t = 5T$. This consideration is 161
 not only true when there is no control on the rotor current, 162
 but also when it is controlled, as shown in the next section. 163

164 VII. DFIG WITH CONTROLLED ROTOR CURRENT UNDER 165 SYMMETRICAL SAGS WITH *Abrupt* RECOVERY

Fig. 4 shows that noticeable rotor current peaks appear in 166
 a DFIG subject to *abrupt* recovery sags if there is no control 167
 on the rotor current. As these peaks cannot be tolerated by the 168
 converter, the DFIG control must avoid or mitigate them. 169

A simple control algorithm is considered in this paper: the 170
 current i_{rf} (rotor current in the synchronous reference frame) 171
 is kept constant at its prefault steady-state value. The control 172
 is, moreover, assumed ideal, which means that the controlled 173
 variable i_{rf} is adjusted instantaneously to satisfy the set 174
 point requirement, i.e., i_{rf} is kept constant during the entire 175

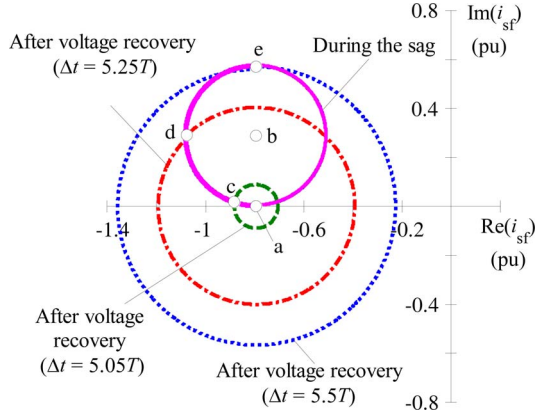


Fig. 5. Dynamic behavior of the DFIG with controlled rotor current. Locus of the transformed stator current under three abrupt recovery sags with different durations ($\Delta t = 5.05T$, $\Delta t = 5.25T$, $\Delta t = 5.5T$). Sags characteristics: $h = 0.1$ and $\psi = 80^\circ$. WT operating point 1: P_n and $s = -0.27$.

event. Although this control strategy is not an optimal control, it allows the electrical transient to be solved analytically (see Appendix II), providing an excellent tool for the comprehension of DFIG dynamic behavior subject to sags.

Throughout this section, it is assumed that the WT operates at point 1 ($P = P_n$ and $s = -0.27$) with unity stator power factor, where the prefault steady-state values are $v_{sf} = 1$ pu, $v_{rf} = -0.27 - j0.04$ pu, $i_{sf} = -0.79$ pu, $i_{rf} = 0.82 - j0.34$ pu, and $p = -1$ pu. The minus sign in i_{sf} and p means that the active power is delivered to the grid, according to the motor sign convention.

A. Transformed Stator Current Analytical Solution

The analytical solution of the transformed stator current i_{sf} before, during, and after the sag is developed in Appendix II and summarized in (23)–(25). The final expressions are

$$\begin{aligned} i_{sf} &= \frac{v_{sf} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s} & (t \leq t_i) \\ i_{sf} &= \underline{K}_1 e^{-(R_s/L_s)(t-t_i)} e^{-j\omega_s(t-t_i)} + \frac{v_{sfh} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s} & (t_i \leq t \leq t_f) \\ i_{sf} &= \underline{K}_2 e^{-(R_s/L_s)(t-t_f)} e^{-j\omega_s(t-t_f)} + \frac{v_{sf} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s} & (t \geq t_f) \end{aligned} \quad (4)$$

where v_{sf} is the transformed stator voltage before the sag, $v_{sfh} = hv_{sf}$ is the transformed stator voltage during the sag, i_{rf} is the steady-state transformed rotor current, t_i is the instant when the sag begins, t_f is the instant when the sag ends, and $\underline{K}_1$ and $\underline{K}_2$ are the following complex constants:

$$\begin{aligned} \underline{K}_1 &= \frac{v_{sf} - v_{sfh}}{R_s + j\omega_s L_s} \\ \underline{K}_2 &= \underline{K}_1 [e^{-(R_s/L_s)(t_f-t_i)} e^{-j\omega_s(t_f-t_i)} - 1]. \end{aligned} \quad (5)$$

Next, the behavior of the DFIG subject to symmetrical sags is analyzed by (4). Fig. 5 shows the locus (real part versus

imaginary part) of the transformed stator current i_{sf} for three different sag durations ($\Delta t = 5.05T$, $\Delta t = 5.25T$, and $\Delta t = 5.5T$). The dynamic behavior of i_{sf} can be explained with the help of points a–e in Fig. 5.

- 1) Point a: the prefault steady-state current i_{sf} is at point a. This current is given by the first equation of (4). The sag begins at t_i and current i_{sf} at this moment equals its prefault steady-state value, $i_{sf}(t_i)$.
- 2) Point b: current i_{sf} at point a, $i_{sf}(t_i)$, evolves into a spiral (magenta solid line in Fig. 5) centered at point b. Current i_{sf} at point b is the steady-state current imposed by the voltage sag v_{sfh} . The damping of the spiral is the exponential term in the second equation of (4). This exponential term contains the pulsation ω_s , which means that the transformed stator current is rotating at ω_s during the sag.
- 3) Point c: if the fault is cleared when i_{sf} is at point c, a small transient occurs in current i_{sf} after fault clearing (green dashed line in Fig. 5). This is because i_{sf} at point c is close to the prefault steady-state value (point a).
- 4) Point d: if the fault is cleared when i_{sf} is at point d, a severe transient occurs in current i_{sf} after fault clearing (red dashed-dotted line in Fig. 5). This is because i_{sf} at point d is far enough from the prefault steady-state value (point a).
- 5) Point e: if the fault is cleared when i_{sf} is at point e, the most severe transient occurs in current i_{sf} after fault clearing (blue dotted line in Fig. 5). This is because i_{sf} at point e is at the farthest point from the prefault steady-state value (point a).

As can be seen in Fig. 5, the most unfavorable voltage recovery instant corresponds to point e, while the most favorable voltage recovery instant corresponds to point c (or to any other point located in the vicinity of point a).

Current i_{sf} evolves into a new spiral after fault clearing, as can be deduced from the third equation of (4), until reaching the prefault steady-state value (point a).

Fig. 6 shows the dynamic behavior of the most relevant DFIG variables (the stator currents and the rotor voltages) for sags with durations $\Delta t = 5T$, $\Delta t = 5.25T$, and $\Delta t = 5.5T$. These cases correspond to the three cases analyzed in Fig. 5.

Fig. 6(a) shows the locus of current i_{sf} . This figure illustrates the behavior described in Fig. 5 again, where the worst situation occurs when sag duration is $\Delta t = 5.5T$ because the transformed stator current $i_{sf}(t_f)$ at the voltage recovery instant is located at the farthest position from the prefault steady-state value, $i_{sf}(t_i)$.

Fig. 6(b) shows the temporal evolution of the transformed stator current i_{sf} and of the abc phase currents i_{sabc} . It is observed that the dynamic behavior of the transformed stator current during the sag is identical for all sag durations. However, the behavior after voltage recovery is different for different sag durations. In the first case, $\Delta t = 5T$, there is a small fluctuation in the real and imaginary parts of current i_{sf} . In the second case, $\Delta t = 5.25T$, large fluctuations occur. The third case, $\Delta t = 5.5T$, exhibits the most severe fluctuations. These dynamic responses are in accordance with the previous remarks in Figs. 5 and 6(a): if sag duration is $\Delta t = 5T$, the sag ends when current

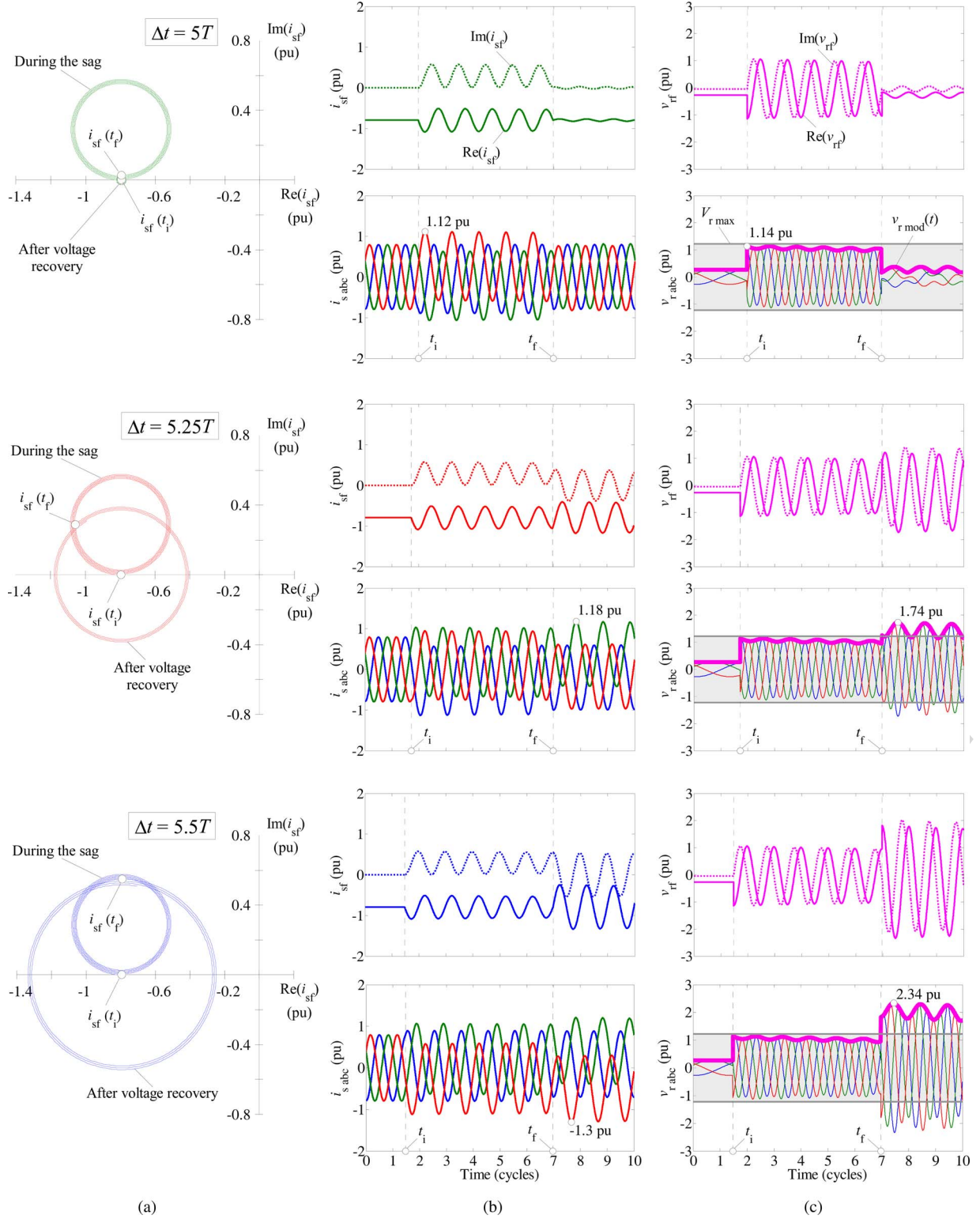


Fig. 6. Influence of sag duration on the dynamic behavior of the DFIG with controlled rotor current. *Abrupt* recovery sag characteristics: $h = 0.1$, $\psi = 80^\circ$, and durations $\Delta t = 5T$, $\Delta t = 5.25T$, and $\Delta t = 5.5T$. WT operating point 1: P_n and $s = -0.27$. (a) Locus of the transformed stator current, (b) transformed and abc phase components of the stator currents, and (c) transformed and abc phase components of the rotor voltages. The shaded area corresponds to the situations where the rotor current can be controlled.

255 $i_{sf}(t_f)$ is close to the prefault steady-state value $i_{sf}(t_i)$. Then,
 256 the fluctuation in the current is small. However, if sag duration
 257 increases, $i_{sf}(t_f)$ is farther from the prefault steady-state value,
 258 causing a large fluctuation.

259 B. Transformed Rotor Voltage Analytical Solution

260 Appendix II also includes the analytical solution of the trans-
 261 formed rotor voltage v_{rf} before, during, and after the sag (26).
 262 It is used in Fig. 6(c) to represent the temporal evolution of the
 263 real and imaginary parts of v_{rf} and of the abc phase components
 264 of the rotor voltages v_{rabc} . This solution assumes that the gen-
 265 erator speed ω_m does not change significantly during the fault
 266 (around 100 ms). The figure also shows the voltage limit V_{rmax}
 267 of the rotor-side converter (defined in Appendix III), which is
 268 obtained by considering that the rotor-side converter power is
 269 100% of the DFIG rated power. This should not pose a prob-
 270 lem as the converter cost only accounts for 5% of the total WT
 271 cost [14]. Therefore, this increase in the cost should be of minor
 272 importance if WT fault ride-through capability is enhanced.

273 Fig. 6(c) also illustrates the modulus of the instantaneous
 274 rotor voltage, defined as

$$\begin{aligned} v_{rmod}(t) &= \frac{2}{\sqrt{3}} |v_{rf}(t)| \\ &= \sqrt{\frac{2}{3}} \sqrt{v_{ra}^2(t) + v_{rb}^2(t) + v_{rc}^2(t)} \\ v_{rmodpu}(t) &= |v_{rfpu}(t)| \\ &= \sqrt{\frac{2}{3}} \sqrt{v_{rapu}^2(t) + v_{rbpu}^2(t) + v_{rcpu}^2(t)} \quad (6) \end{aligned}$$

275 where $v_{ra}(t)$, $v_{rb}(t)$, and $v_{rc}(t)$ are the instantaneous rotor phase
 276 voltages. The pu values of $v_{rf}(t)$, $v_{ra}(t)$, $v_{rb}(t)$, and $v_{rc}(t)$ are
 277 calculated according to (3), and the base value for $v_{rmod}(t)$ is
 278 $\sqrt{2}V_b$. The usefulness of $v_{rmod}(t)$ is that it represents the rotor
 279 voltage required to maintain the current i_{rf} constant, i.e., when
 280 $v_{rmod}(t)$ satisfies

$$v_{rmod}(t) \leq V_{rmax} \quad (7)$$

281 the current i_{rf} can be kept constant or, in general, the rotor
 282 current can be controlled. For brevity, the modulus $v_{rmod}(t)$ of
 283 the instantaneous rotor voltage is shortened as *mod* rotor voltage
 284 in the remainder of this paper.

285 Fig. 6(c) shows that (7) is satisfied during the sag, but it
 286 is not always satisfied after fault clearing. Current i_{rf} can be
 287 controlled and kept constant if $\Delta t = 5T$. However, if sag dura-
 288 tion increases, current i_{rf} cannot be controlled after fault clear-
 289 ing because the rotor-side converter requires a higher voltage
 290 than its voltage limit, $\Delta t = 5.5T$ being the most unfavorable
 291 duration.

292 C. (Abrupt Recovery) Sag Duration Periodicity Influence

293 *Abrupt* recovery sags with durations different from those
 294 shown in this paper, but still periodic durations, produce quasi-
 295 identical effects because (4) is poorly damped. Then, the most
 296 unfavorable (*abrupt* recovery) sag durations (with the rotor cur-

TABLE II
DISCRETE SAG FAULT-CLEARING INSTANTS (ADAPTED FROM [4])

Type	First recovery (ωt_{f1})	Second recovery (ωt_{f2})	Third recovery (ωt_{f3})
A ₁	$n 360^\circ - \alpha_a + \psi$	$n 360^\circ - \alpha_a + \psi + 90^\circ$	—
A ₂	$n 360^\circ - \alpha_a + \psi + 90^\circ$	$n 360^\circ - \alpha_a + \psi + 180^\circ$	—
A ₃	$n 360^\circ - \alpha_a + \psi$	$n 360^\circ - \alpha_a + \psi + 60^\circ$	$n 360^\circ - \alpha_a + \psi + 120^\circ$
A ₄	$n 360^\circ - \alpha_a + \psi + 90^\circ$	$n 360^\circ - \alpha_a + \psi + 150^\circ$	$n 360^\circ - \alpha_a + \psi + 210^\circ$
A ₅	$n 360^\circ - \alpha_a + \psi$	$n 360^\circ - \alpha_a + \psi + 60^\circ$	$n 360^\circ - \alpha_a + \psi + 120^\circ$

Notes: - $n = 0, 1, 2, \dots$
 - α_a = phase angle of phase-a voltage
 - ψ = fault current angle (equal for all sag types)

rent control used in this study) can be written in the form $\Delta t =$
 $nT + 0.5T$, where $n = 0, 1, 2, \dots$. In the authors' knowledge,
 the duration periodicity effect is also produced when there is no
 control on the rotor current.

VIII. DFIG UNDER SYMMETRICAL SAGS WITH Discrete RECOVERY

In Sections VI and VII, it is assumed that faults are cleared
 instantaneously in all affected phases, i.e., sags are modeled
abrupt. The *abrupt* symmetrical sag is denoted as type A. In
 the remaining sections of this paper, it is assumed that faults are
 cleared in one, two, or three steps, i.e., sags are modeled *discrete*.
 There are different ways to fully clear the same type of fault.
 As far as symmetrical sags are concerned, these possibilities are
 classified into five cases [4], denoted as A₁, A₂, A₃, A₄, and A₅,
 which are shown in Table II (adapted from [4]). Fig. 2 shows
 the rms voltage waveform pattern of a type A (*abrupt*) sag and
 a type A₃ (*discrete*) sag.

It should be noted that types A₃ and A₅ only differ in the
homopolar-sequence components, which means that their ef-
 fects are identical if the fault occurs at a different voltage level
 from which the DFIG is connected to, or if the DFIG is un-
 grounded. Since DFIG windings are ungrounded (they are iso-
 lated wye or delta connected), type A₃ is chosen. Furthermore,
 sags with two-step voltage recovery (types A₁ and A₂) cause
 similar effects on DFIG behavior and the same is true for types
 A₃ and A₄ (three-step recovery sags). Thus, only types A₁ and
 A₃ are considered to study such effects.

In order to clarify the *discrete* recovery effects on DFIG be-
 havior, Fig. 7(a) shows the locus (real part versus imaginary
 part) of the transformed stator current i_{sf} , considering *discrete*
 recovery sag types A₁ and A₃. The sag duration in this figure
 is the most unfavorable duration for *abrupt* recovery sags, i.e.,
 $\Delta t = 5.5T$. The WT operates at point 1 (P_n and $s = -0.27$).
 The behavior in Fig. 7 is similar to that upon occurrence of
abrupt recovery sags (see Fig. 6): the current evolves into a
 damped spiral when the fault begins and develops into another
 damped spiral when the fault is effectively cleared. However, as
 the fault is cleared in two or three steps, the development from
 the first to the second spiral is not *abrupt*, but in two or three
 steps, respectively. This causes the second spiral to be smaller in
 a *discrete* recovery sag [see Fig. 7(a)] than in an *abrupt* recov-
 ery sag [see Fig. 6(a)]. As a result, the stator current peaks [see
 Fig. 7(b)] and the rotor voltage peaks [see Fig. 7(c)] are smaller

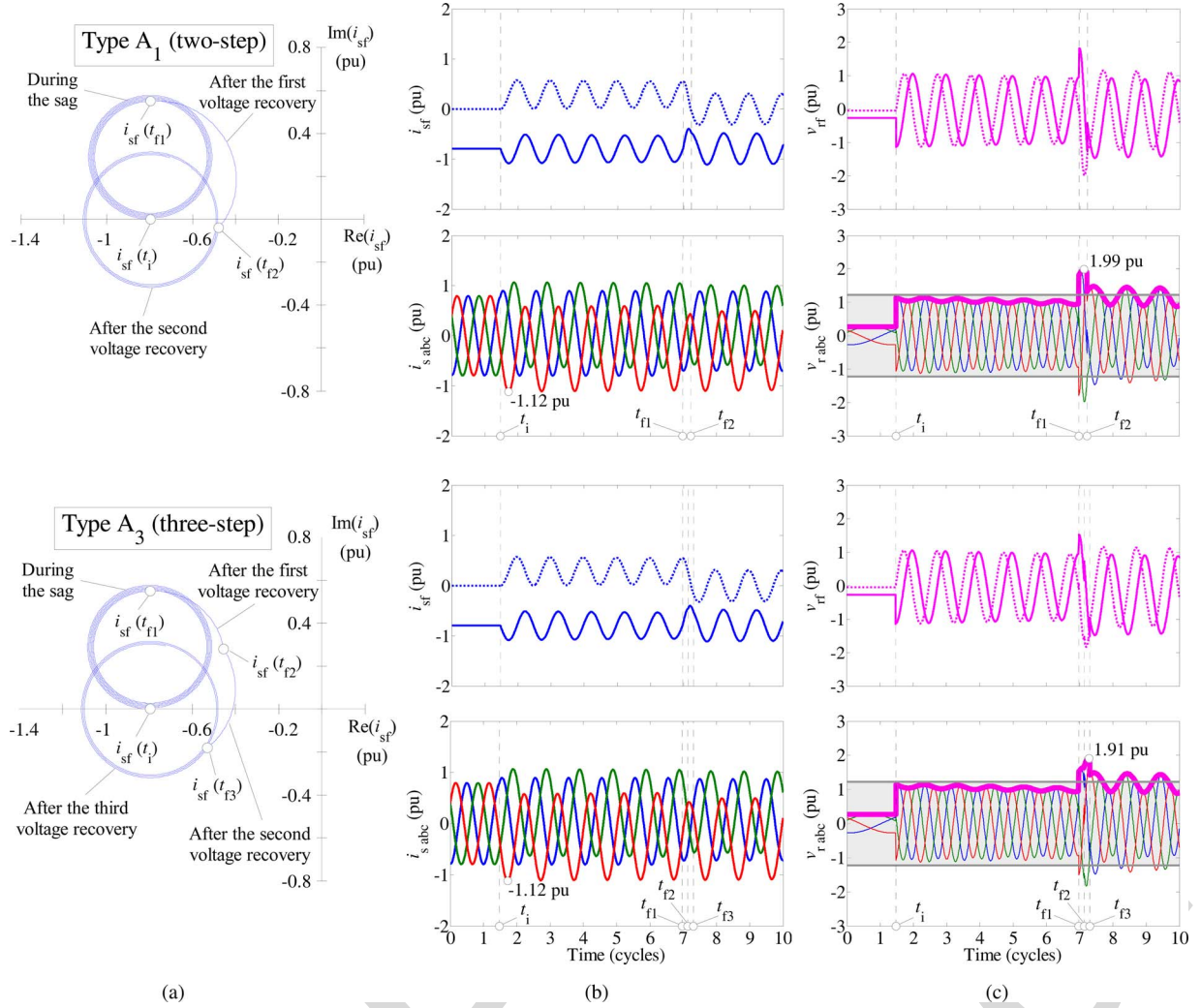


Fig. 7. Depth influence on DFIG behavior. *Abrupt* recovery sags (type A) and *discrete* recovery sags (types A₁ and A₃). Sags characteristics: $h = 0.1$, $\Delta t = 5.5T$, and $\psi = 80^\circ$. (a) Stator current peak, (b) *mod* rotor voltage peak, and (c) mean rotor voltage. WT operating points: 1 (P_n , solid line), 2 ($0.5P_n$, dashed line), and 3 ($0.1P_n$, dotted line). The shaded area corresponds to the situations where the rotor current can be controlled.

than for the *abrupt* recovery sag. To obtain v_{rf} and $v_{r \text{ abc}}$ in Fig. 7(c), it is assumed that the generator speed ω_m does not change significantly during the fault (around 100 ms).

These examples suggest that the increase in the number of recovery steps smoothes the sag effects. In fact, the maximum value of the rotor voltage is lower in the type A₃ sag (three-step recovery sag) than in the type A₂ sag (two-step recovery sag), and this effect is always repeated for all the three WT operating points.

As concluded in the previous section, Fig. 6 shows that the rotor current cannot be controlled for a deep ($h = 0.1$) type A sag, i.e., for a deep sag with *abrupt* recovery, because the rotor voltage required to maintain constant the transformed rotor current is higher than the rotor-side converter voltage limit, $V_{r \text{ max}}$. On the contrary, Fig. 7 illustrates that the rotor current can be controlled if the same deep ($h = 0.1$) sag is modeled with *discrete* recovery, i.e., types A₁ and A₃, because the required rotor voltage fluctuates around $V_{r \text{ max}}$ after fault clearing. Strictly speaking, there is a short time interval during fault clearing where rotor current cannot be controlled as a peak higher than

$V_{r \text{ max}}$ appears in the required rotor voltage. Note also that the stator current peaks are always less pronounced for the *discrete* recovery cases than for the *abrupt* recovery case.

Thus, this example demonstrates that the severe effects of *abrupt* recovery sags are less pronounced when the fault-clearing physics is considered, i.e., when the sag is modeled *discrete*.

IX. SITUATIONS WHERE THE ROTOR CURRENT CAN BE CONTROLLED IN BOTH *Abrupt* AND *Discrete* SAGS

The peaks of the stator current and the peaks of the modulus of the instantaneous rotor voltage can be chosen to compare the effects of both *abrupt* and *discrete* symmetrical sags on the DFIG behavior. These peak values are referred to the base values ($S_b = 2 \text{ MW}$, $U_b = 690 \text{ V}$, $I_b = S_b / (\sqrt{3}U_b) = 1673.5 \text{ A}$)

$$i_{s \text{ peak pu}} = \frac{i_{s \text{ peak}}}{(\sqrt{2}I_b)} = \frac{\max\{|i_{sa}(t)|, |i_{sb}(t)|, |i_{sc}(t)|\}}{(\sqrt{2}I_b)}$$

$$v_{r \text{ mod peak pu}} = \frac{v_{r \text{ mod peak}}}{(\sqrt{2}V_b)} = \frac{\max\{v_{r \text{ mod}}(t)\}}{(\sqrt{2}V_b)} \quad (8)$$

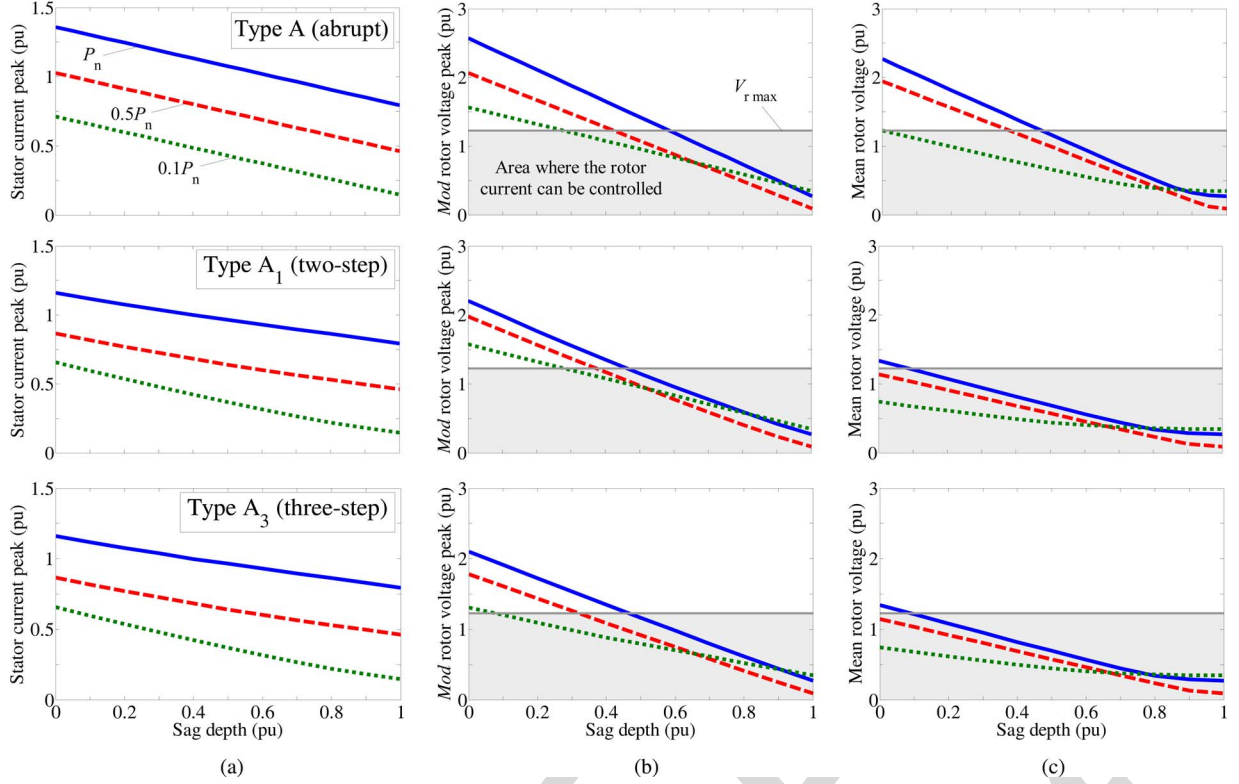


Fig. 8. Depth influence on DFIG behavior. *Abrupt* recovery sags (type A) and *discrete* recovery sags (types A₁ and A₃). Sags characteristics: $h = 0.1$, $\Delta t = 5.5T$ and $\psi = 80^\circ$. (a) Stator current peak, (b) *mod* rotor voltage peak, and (c) mean rotor voltage. WT operating points: 1 (P_n , solid line), 2 ($0.5P_n$, dashed line), and 3 ($0.1P_n$, dotted line). The shaded area corresponds to the situations where the rotor current can be controlled.

where $i_{sa}(t)$, $i_{sb}(t)$, and $i_{sc}(t)$ are the abc phase components of the stator currents, and $v_{r\text{mod}}(t)$ is the modulus of the abc instantaneous rotor voltage (*mod* rotor voltage), defined in (6).

Furthermore, in order to analyze if the rotor current can be controlled after fault clearing, the mean value of the *mod* rotor voltage is chosen as another variable to characterize DFIG behavior. This mean value is defined as follows:

$$v_{r\text{mean}} = \frac{1}{T} \int_{t_1}^{t_1+T} v_{r\text{mod}}(t) dt \rightarrow v_{r\text{mean pu}} = \frac{v_{r\text{mean}}}{(\sqrt{2}V_b)}. \quad (9)$$

The mean value of the *mod* rotor voltage is evaluated at the first period after the *mod* rotor voltage is maximum. Instant t_1 is chosen as $t_1 = t_{vr\text{max}} + T/2$ ($t_{vr\text{max}}$ is the instant when the *mod* rotor voltage is maximum). This mean value is shortened as mean rotor voltage in the remainder of this paper.

Fig. 8 illustrates the influence of sag depth h on the stator current peak, the *mod* rotor voltage peak, and the mean rotor voltage for types A, A₁, and A₃ for the most unfavorable duration according to the previous results, i.e., $\Delta t = 5.5T$.¹ In this figure, the three WT operating points illustrated in Fig. 3 are considered, which correspond to different generated powers.

Fig. 8(b) shows that the rotor current in types A₁ and A₃ can be controlled (it can also be controlled during the fault-clearing

instants) if $h > 0.45$, while for *abrupt* recovery sags it is only accomplished when $h > 0.6$. In these situations, the peak value of $v_{r\text{mod}}(t)$ is always below the rotor-side converter voltage limit, $V_{r\text{max}}$. Apart, it is observed that if the WT operates at different points, these limits are different. For example, when the sag is *abrupt* and WT operates at points 2 and 3, the rotor current can be controlled if $h > 0.4$ and $h > 0.3$, respectively, while if a three-step recovery sag is considered (type A₃), these limits are extended $h > 0.3$ for WT operating point 2 and $h > 0.1$ for point 3.

Moreover, Fig. 8(c) confirms that in *discrete* recovery sags $v_{r\text{mean}}$ is lower than in *abrupt* recovery sags. Thus, the rotor current can be controlled for *discrete* recovery sags with $h > 0.1$ (for any WT operating point) except for the short time interval during fault clearing when a peak higher than $V_{r\text{max}}$ appears in $v_{r\text{mod}}(t)$. In contrast, in *abrupt* cases the rotor current can only be controlled if $h > 0.5$.

Then, it can be concluded that *discrete* recovery sags are less severe than *abrupt* recovery sags, because they let more possible situations where the rotor current can be controlled. Apart, the WT generated power also has an important influence on these situations, being the WT operating point 1 the most unfavorable one, and point 3 the most favorable one.

X. CONCLUSION

An analytical study of the dynamic behavior of the DFIG subject to symmetrical voltage sags is presented. This study helps understand WT fault ride-through capability.

¹Extensive simulations not included in this paper demonstrate that (with the rotor current control used in this work) the most unfavorable durations for types A₁ and A₃ are $\Delta t = 5.7T$ and $\Delta t = 5.6T$, respectively. However, the differences between considering $\Delta t = 5.5T$ and $\Delta t = 5.7T$ for type A₁, and $\Delta t = 5.5T$ and $\Delta t = 5.6T$ for type A₃, are less than 2%.

The assumption that the rotor current i_{rf} in the synchronous reference frame is kept constant makes it possible to obtain an analytical expression for the dynamic stator currents and required rotor voltages. The analytical approach provides a comprehensive description of sag effects on the stator currents and rotor voltages.

The study shows that sag effects on DFIG behavior are periodic with sag duration. Furthermore, there is a duration which produces the most severe effects (in the idealized rotor current control used in this paper, the most severe effects on DFIG behavior can be written in the form $\Delta t = nT + mT$, $n = 0, 1, 2, \dots$, and $m = 0.5, 0.7$, or 0.6 for types A, A_1 and A_3 , respectively).

Sag modeling and its influence on the estimation of sag event effects are also discussed in this paper. When a sag is modeled *abrupt*, the rotor voltages required after fault clearing to maintain a constant current i_{rf} are higher than the rotor-side converter voltage limit. Consequently, this current cannot be controlled. However, this effect is smoothed if fault-clearing physics is considered, i.e., if sag is modeled *discrete*. This is because the successive voltage recovery steps smooth the rotor voltage effects. As a consequence, the rotor current in almost all *discrete* sags can be controlled once the fault is cleared. As the *discrete* recovery sag is a more accurate model than the *abrupt* one (which is the usual modeling in the technical literature), it can be concluded that *abrupt* models overestimate the severity of sag events.

In conclusion, it is important to consider the following three items when analyzing other nonidealized controls for the DFIG rotor current: 1) to study the possible duration periodicity effect; 2) to find the sag duration that produces the most severe effects; and 3) to consider a *discrete* model for the sag.

APPENDIX I

RELATION BETWEEN ABC PHASE COMPONENTS AND TRANSFORMED COMPONENTS

The original Ku transformation is defined in [11] and [12]. In this paper, it is used in the normalized or power-invariant form

$$[K(\Psi)] = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & e^{j\Psi} & e^{-j\Psi} \\ 1 & a^2 e^{j\Psi} & a e^{-j\Psi} \\ 1 & a e^{j\Psi} & a^2 e^{-j\Psi} \end{bmatrix} \quad (10)$$

where $a = e^{j2\pi/3}$ and Ψ is the transformation angle. This matrix relates the abc phase components of a three-phase system to the Ku transformed components, namely, *homopolar* (0), *forward* (f), and *backward* (b). In the case of the induction machine, the transformation angle for the stator variables is Ψ_s , while the transformation angle for the rotor variables is $\Psi_r = \Psi_s - \varphi\theta_m$, where θ_m is the rotor mechanical angle. Then

$$\begin{aligned} [x_{sabc}] &= [K(\Psi_s)][x_{s0fb}] \\ [x_{rabc}] &= [K(\Psi_s - \varphi\theta_m)][x_{r0fb}] \end{aligned} \quad (11)$$

$$\begin{aligned} [x_{s0fb}] &= [K(\Psi_s)]^{-1}[x_{sabc}] \\ [x_{r0fb}] &= [K(\Psi_s - \varphi\theta_m)]^{-1}[x_{rabc}] \end{aligned} \quad (12)$$

where x stands for the variable of interest (voltage, current, or flux). The stator transformation angle in the synchronous reference frame is $\Psi_s = \omega_s t$. Note that when the rotor mechanical speed is constant, the rotor transformation angle is simplified as $\Psi_r = \Psi_s - \varphi\theta_m = \omega_s t - (\varphi\omega_m t + \varphi\theta_{m0}) = s\omega_s t - \varphi\theta_{m0}$, being θ_{m0} the rotor mechanical angle at $t = 0$.

To study the DFIG behavior by means of the transformed Ku variables, two considerations are important. First, no *homopolar* components are present because the stator and rotor windings are isolated wye or delta connected. Second, as the *backward* components equal the complex conjugate of the *forward* components, only the *forward* components are studied [12]. Thus, the abc phase stator and rotor variables (currents, voltages, or fluxes) are obtained by applying (11)

$$\begin{aligned} x_{sa} &= \frac{2}{\sqrt{3}} \text{Re}(e^{j\omega_s t} x_{sf}), & x_{ra} &= \frac{2}{\sqrt{3}} \text{Re}(e^{j(\omega_s t - \varphi\theta_m)} x_{rf}) \\ x_{sb} &= \frac{2}{\sqrt{3}} \text{Re}(a^2 e^{j\omega_s t} x_{sf}), & x_{rb} &= \frac{2}{\sqrt{3}} \text{Re}(a^2 e^{j(\omega_s t - \varphi\theta_m)} x_{rf}) \\ x_{sc} &= \frac{2}{\sqrt{3}} \text{Re}(a e^{j\omega_s t} x_{sf}), & x_{rc} &= \frac{2}{\sqrt{3}} \text{Re}(a e^{j(\omega_s t - \varphi\theta_m)} x_{rf}). \end{aligned} \quad (13)$$

On the other hand, the transformed Ku variables (*forward* component) are obtained by applying (12)

$$\begin{aligned} x_{sf} &= \frac{e^{-j\omega_s t}}{\sqrt{3}} (x_{sa} + ax_{sb} + a^2 x_{sc}) \\ x_{rf} &= \frac{e^{-j(\omega_s t - \varphi\theta_m)}}{\sqrt{3}} (x_{ra} + ax_{rb} + a^2 x_{rc}). \end{aligned} \quad (14)$$

The *forward* component of the stator voltages is

$$v_{sf} = \frac{e^{-j\omega_s t}}{\sqrt{3}} (v_{sa} + av_{sb} + a^2 v_{sc}) \quad (15)$$

where v_{sa} , v_{sb} , and v_{sc} are the abc stator per-phase voltages. When these voltages are sinusoidal and form a set of balanced voltages (as is the case of the prefault voltages), v_{sf} is also related to the stator per-phase voltage phasor $\underline{V}_s$ just as the sinusoidal steady-state values of i_{sf} , v_{rf} , and i_{rf} are related to the corresponding phasors $\underline{I}_s$, $\underline{V}_r$, and $\underline{I}_r$

$$v_{sf} = \sqrt{\frac{3}{2}} \underline{V}_s \quad i_{sf} = \sqrt{\frac{3}{2}} \underline{I}_s \quad v_{rf} = \sqrt{\frac{3}{2}} \underline{V}_r \quad i_{rf} = \sqrt{\frac{3}{2}} \underline{I}_r. \quad (16)$$

The Ku *forward* component is related to the *Park dq* components as follows [12]:

$$x_d = \sqrt{2} \text{Re}(x_f) \quad x_q = \sqrt{2} \text{Im}(x_f) \quad (17)$$

where subscripts d and q stand for the *direct* and *quadrature* components of the *Park* transformation, respectively.

APPENDIX II

ANALYTICAL STUDY OF A DFIG UNDER

Abrupt SYMMETRICAL SAGS

A. Transformed Stator Current Analytical Solution

Consider that the control imposes a constant rotor current i_{rf} in the synchronous reference frame; then, $p i_{rf} = 0$ in (1), which results in

$$\begin{aligned} v_{sf} &= [R_s + L_s (p + j\omega_s)] i_{sf} + j\omega_s M i_{rf} \\ v_{rf} &= (R_r + j\omega_s L_r) i_{rf} + M (p + j\omega_s) i_{sf}. \end{aligned} \quad (18)$$

The current i_{sf} is obtained from the first equation of (18)

$$p i_{sf} = \frac{1}{L_s} [v_{sf} - (R_s + j\omega_s L_s) i_{sf} - j\omega_s M i_{rf}] \quad (19)$$

which is a first-order linear differential equation with constant coefficients and complex variables. Its analytical solution can be obtained as the sum of the homogeneous and particular solutions [15]. The homogeneous solution $i_{sf \text{ homog}}$ is the solution of (19) with $v_{sf} = 0$ and $i_{rf} = 0$, while the particular solution $i_{sf \text{ part}}$ is the steady-state solution of (19), i.e., the solution of (19) with $p i_{sf} = 0$. Then, the complete solution of (19) results in

$$\begin{aligned} i_{sf} &= i_{sf \text{ homog}} + i_{sf \text{ part}} \\ &= \underline{K} e^{-(R_s/L_s)t} e^{-j\omega_s t} + \frac{v_{sf} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s}. \end{aligned} \quad (20)$$

The complex constant $\underline{K}$ is calculated by considering the initial condition $i_{sf}(t_0) = i_{sf0}$ for $t = t_0$, resulting in

$$\underline{K} = \left(i_{sf0} - \frac{v_{sf} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s} \right) e^{(R_s/L_s)t_0} e^{j\omega_s t_0}. \quad (21)$$

By replacing (21) in (20) and grouping terms, we have

$$\begin{aligned} i_{sf} &= \left(i_{sf0} - \frac{v_{sf} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s} \right) e^{-(R_s/L_s)(t-t_0)} e^{-j\omega_s(t-t_0)} \\ &\quad + \frac{v_{sf} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s} \end{aligned} \quad (22)$$

which is the solution of differential equation (19), i.e., the *forward* component of the DFIG stator current when the current i_{rf} is kept constant. This equation is valid for the three states of an *abrupt* sag event: before the sag, during the sag, and after *abrupt* voltage recovery. Terms i_{sf0} , t_0 , and v_{sf} of (22) are different for the three states.

1) *State 1: Before the Sag* ($t \leq t_i$): The DFIG is assumed to be in electrical and mechanical steady-state conditions before the sag begins (at $t = t_i$). The transformed stator current can be calculated from the steady-state expression of (22)

$$i_{sf} = \frac{v_{sf} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s} \quad (t \leq t_i). \quad (23)$$

2) *State 2: During the Sag* ($t_i \leq t \leq t_f$): The sag begins at t_i , which is the initial instant t_0 for State 2. During this state, the *forward* component of the stator voltage is $h v_{sf}$, which is renamed v_{sfh} to simplify the notation. The initial current

i_{sf0} is the steady-state current (23). By introducing these three conditions in (22), we have

$$\begin{aligned} i_{sf} &= \left(\frac{v_{sf} - v_{sfh}}{R_s + j\omega_s L_s} \right) e^{-(R_s/L_s)(t-t_i)} e^{-j\omega_s(t-t_i)} \\ &\quad + \frac{v_{sfh} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s} \quad (t_i \leq t \leq t_f). \end{aligned} \quad (24)$$

3) *State 3: Abrupt Voltage Recovery* ($t \geq t_f$): The sag recovers instantaneously (at the same time in the three phases) at t_f , which is the initial instant t_0 for State 3. From this instant, the stator voltage v_{sf} remains at the prefault steady-state value. The initial current i_{sf0} for State 3 is the stator current of (24) evaluated at instant t_f . By introducing these three conditions in (22), we have

$$\begin{aligned} i_{sf} &= \left(\frac{v_{sf} - v_{sfh}}{R_s + j\omega_s L_s} \right) [e^{-(R_s/L_s)(t_f-t_i)} e^{-j\omega_s(t_f-t_i)} - 1] \\ &\quad \times [e^{-(R_s/L_s)(t-t_f)} e^{-j\omega_s(t-t_f)}] + \frac{v_{sf} - j\omega_s M i_{rf}}{R_s + j\omega_s L_s} \quad (t \geq t_f). \end{aligned} \quad (25)$$

B. Transformed Rotor Voltage Analytical Solution

By replacing (19) in the second equation of (18), we obtain

$$\begin{aligned} v_{rf} &= \left(R_r + j\omega_s L_r - \frac{j\omega_s M^2}{L_s} \right) i_{rf} + \frac{M}{L_s} v_{sf} \\ &\quad + \left[j\omega_s M - \frac{M}{L_s} (R_s + j\omega_s L_s) \right] i_{sf} \end{aligned} \quad (26)$$

which is the analytical solution of the *forward* component of the rotor voltages. This expression provides the required rotor voltage to maintain the current i_{rf} constant.

The analytical expressions were validated with numerical models of the DFIG developed with MATLAB and PSpice.

It should be noted that the analytical expressions are valid even if the generator speed ω_m varies during the fault. Note that, unlike i_{sf} , v_{rf} depends on ω_m as it depends on the mechanical slip $s = (\omega_s - \omega_m)/\omega_s$.

C. Analytical Equations for Discrete Recovery Sags

The previous DFIG equations correspond to the *abrupt* recovery sag model. The respective DFIG equations for the *discrete* recovery sag model are not included for brevity but can be obtained in a similar way bearing in mind that types A₁ and A₃ sag events include four and five different states, respectively. Also, the stator voltage between the instants when the first and last phases of the circuit breaker clear the fault (i.e., between the instants t_{f1} and t_{f2} in type A₁, and t_{f1} and t_{f3} in type A₂, see Table II and Fig. 2) is unsymmetrical. This means that voltage v_{sf} is not constant, but of the form $v_{sf} = v_{sf1} + v_{sf2} e^{-j2\omega_s t}$.

APPENDIX III

ROTOR-SIDE CONVERTER VOLTAGE LIMIT

The maximum amplitude of the per-phase fundamental voltage that can be generated by the rotor-side converter is [3]

$$V_{r\max} = m \frac{V_{dc}}{2} \quad (27)$$

where m is the modulation index and V_{dc} is the dc-link voltage. Considering the space vector modulation technique ($m = 1.15$) and a dc-link voltage of 1200 V, a maximum amplitude of $V_{r\max} = 690$ V or 1.22 pu is obtained. It should be remarked that the rotor-side converter power is assumed to be 100% of DFIG rated power. This is an acceptable assumption as the converter cost is around 5% of the total WT system cost [14].

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